



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

B. Sc. CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	INTRODUCTION TO INDUSTRIAL CHEMISTRY				
Type of Course	VOCATIONAL MINOR				
Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Basic awareness on types of industries 2. Types of redox reactions and titrations				
Course Summary	The course explores various industries and their scope, industrial processes, IPR, and analytical knowledge acquisition through volumetric analysis.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the elementary ideas of Cosmetics, Soaps, Detergents, Textiles, Food and Petrochemical industries	U	F	Instructor-created exams / Group Tutorial Work
CO2	Acquire knowledge on elementary concepts of Paints and coatings, Polymers, Fine chemicals and Pharmaceuticals	U	F	Instructor-created exams / Assignments
CO3	Remember the early chemical technologies leading to the Industrial Revolution	R	F	Group Tutorial Work/Seminar
CO4	Analyse various unit processes involved in industry	An	P	Instructor-created exams / Viva
CO5	Acquire skills in various titrimetric analysis of industrial importance	Ap	P	Practical Assignment / Observation of Practical Skills
CO6	Understand the basic concepts, and ethics of Intellectual Property Rights	U	C	Instructor-created exams/Group Tutorial Work
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)				

- Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Industrial chemistry: An Overview		9	20
	1	History of industrial chemistry-Social Background: Industrial Revolution, Early Chemical Technologies.	2	
	2	Types of chemical industries – Glass, Cement, Ceramics, Metals, Steel (Elementary idea only)	1	
	3	Paints and coatings, Polymers, Agrochemicals, Fine chemicals, Pharmaceuticals (Elementary idea only)	1	
	4	Cosmetics, Soaps and Detergents, Textiles, Petrochemicals, Food, Spices, Sugar, Brewery (Elementary idea only)	1	
	5	Management of lab chemicals and equipment, Chemical waste management, fire and health safety management	2	
	6	Methods for procuring chemicals and equipment, record keeping, supply requirements, evaluation of supply trends	2	
II	Industrial Process		12	26
	7	Unit process, unit operations, flow diagrams, Energy balance and materials balance, fuels, calorific value	4	
	8	Fluid flow, streamline flow, turbulence flow, viscosity, Newtonian and non-Newtonian fluids	4	
	9	Heat transfer, types of heat exchangers, refrigeration cycles	4	
III	Intellectual Property Rights (IPR)		9	20
	10	Basic concepts, Purpose and Ethics of IPR	1	
	11	Seven Types including – Copyright & trademarks;	2	
	12	Patents, Geographical indications & Plant varieties	1	
	13	Industrial designs, Layout designs of ICs	2	
	14	Basics of patenting, IP filing methods.	3	

IV	Scope of Industrial Chemistry		15	32
	15	Types of industries – Cottage industries, Small and Large scale industries	2	
	16	Fundamentals of entrepreneurship – Types of entrepreneurs, Entrepreneurship and economic growth	2	
	17	Process of starting a business – Search for business ideas, Sources of business idea, Idea processing, Input requirements	2	
	18	Basics of fund management, fundamentals of finance and budgeting.	2	
	19	Business proposals- Basics of product value assessment, evaluation of product quality testing, evaluation for commercialization.	2	
	20	Basics of overall project management, report preparation and presentation	2	
	21	Start-up and Financial support schemes, MSME	2	
	22	Employment Opportunities – Job positions and eligibility requirement	1	
V	Volumetric Estimation II		30	
	1	Iodimetry and Iodometry 1. Estimation of copper 2. Estimation of arsenious oxide	5	
	2	Complexometric Titrations Using EDTA 3. Estimation of Zn 4. Estimation of Mg	10	
	3	Industrial applications 1. Estimation of Acetic acid content in commercial vinegar 2. Estimation of Alkali content in Antacid Tablets 3. Determination of hardness of water 4. Estimation of Ascorbic acid in fruit juices	15	
Books and References:				

1. B.K Sharma, Industrial Chemistry, Goel Publications (1983).
2. R.K. Das, Industrial Chemistry, Kalyani Publications, New Delhi (1982).
3. W.L.Badger and J.T.Bachero, Introduction to Chemical Engineering, Tata McGraw Hill, U.S.A
4. W.L.McCabe and J.C.Smith, Unit operations in Chemical Engineering, Tata McGraw Hill N.Y
5. J.H.Perry, Chemical Engineering Hand Book, McGraw Hill, N.Y. 4. D.D.Kale, Unit Operations– 1 and
Pune Vidyarthi GrihaPrakashan, Pune
6. K.A.Gavhane, Unit Operations-II Heat and Mass transfer, Nirali Prakashan.
7. A. I. Vogel ‘A Text Book of Quantitative Inorganic Analysis Including Elementary Instrumental Analysis’: (Third Ed.) (ELBS)
8. D.A.Skoog, D.M.West and S.R.crouch, Fundamentals of Analytical Chemistry, 8 thEdn., Brooks/Cole
Nelson

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 2	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 3	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 4	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 5	-	-	2	-	-	2	3	-	2	-	3	-	3
CO 6	-	-	-	-	-	2	3	-	2	-	3	-	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment / Discussion / Seminar
- Internal Examination
- Practical Examination
- Final Examination (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3		✓		✓
CO 4	✓	✓		✓
CO 5		✓	✓	
CO 6	✓			✓